

LAB 3 · ARCHITECTURAL PATTERN SELECTION

Warehouse Inventory & Pallet Location Tracker

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Architecture Selection: We chose Layered Architecture for the Warehouse Inventory & Pallet Location Tracker.

Architectural style analysis

Style	Strength in this scenario	Trade-off	Fit
Layered	Clear separation of UI, business rules and persistence.	Minor layer-call overhead.	Best
Microservices	Independent deployment and fault isolation.	Network, deployment and data-consistency complexity is excessive here.	Low
Client-Server	Central data control and simple clients.	Server/network dependency risks floor operations and latency.	Medium

Two scenario-specific reasons

- **1. Cohesive business rules.** Pallet placement must identify a tag, validate rack load, save a complete 3D location and log the movement. Keeping these rules in a business layer prevents the operator UI and device adapter from duplicating safety-critical logic.
- **2. Maintainability and replaceability.** The touch-screen UI, barcode/Rfid adapter and database can change behind stable interfaces. A scanner vendor, storage engine or screen layout can therefore be replaced without rewriting the capacity-validation rules.

Security advantage and performance benefit

Security advantage All inventory writes pass through the Inventory & Pallet Manager, which can enforce authentication and role-based authorization before database access. The UI and scanner adapter receive no direct database credentials. Parameterised repository queries and an append-only movement audit trail further protect integrity and accountability.	Performance benefit The business layer can validate a rack and persist the placement in one local workflow, avoiding remote microservice calls. Indexed SKU/barcode/Rfid and 3D-location fields, prepared queries, connection pooling and caching of rack limits support the Lab 1 target of returning coordinates across 100,000 bins in under 100 ms.
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Selected components and interfaces

Components: Warehouse Operator UI; Inventory & Pallet Manager; Rack Capacity Service; Barcode/Rfid Adapter; Warehouse Database. **Interfaces:** IWarehouseOperations, IScanData, ICapacityValidation and IWarehouseRepository. The component diagram uses ball-and-socket assembly connectors and labels the protocol/data carried by each interaction.